Name: _____ Period: ____

Quiz Review: Sections 2.1–2.3

Honors Physics · Unit 2 · Scalars, vectors, resultants, and components

Show your work: sketch each vector or triangle, write the equation you use, substitute with units, and box your answer. East and north are positive. Give directions as an angle from a reference direction, such as 37° N of E.

$A_x = A\cos\theta \qquad A_y = A\sin\theta \qquad A = \sqrt{A_x^2 + A_y^2} \qquad \theta = \tan^{-1}\left(\frac{|A_y|}{|A_x|}\right)$

1. Mr. Peck runs three sides of a rectangular park. Starting at the southwest corner, Mr. Peck runs 400 m east, then 300 m north, then 400 m west. The whole run takes 6.0 min.

a) Find the distance traveled and the displacement (magnitude and direction).

b) Find the average speed and the average velocity.

c) At which corner is Mr. Peck farthest from the start? How far, and in what direction?

d) Mr. Peck's watch shows an average of 3.06 m/s, but a GPS tracker reports a much smaller average velocity. Explain how both can be correct.

2. Two tow straps pull on a car stuck in the mud. One strap pulls with 650 N and the other with 250 N.

a) What are the largest and smallest possible magnitudes of the total pull? Sketch each case tip-to-tail. Could the total pull be 300 N?

b) The straps pull at right angles: 650 N east and 250 N north. Find the magnitude and direction of the total pull.

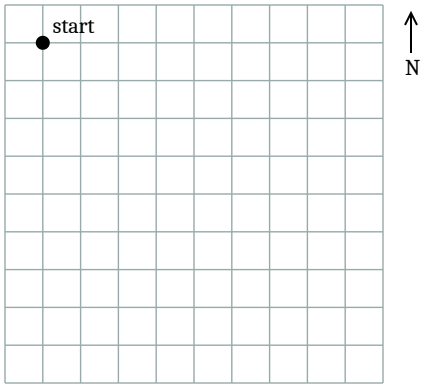
3. Mr. Jantz follows geocache clues: walk 60 m east, then 80 m south, then 20 m west. On the grid, each square is 10 m and the dot is the start.

a) Draw the three moves tip-to-tail and draw the resultant. Measure its length and angle.

b) Calculate the magnitude and direction of the resultant.

c) How much farther did Mr. Jantz walk than the straight-line distance to the cache?

d) If the clues were done in a different order, would the cache location change? Would the distance walked change?



1 square = 10 m

4. **Challenge.** Mrs. Luksic throws a softball from deep center field at 27.0 m/s. The horizontal component of its velocity is 24.0 m/s.

Find the launch angle and the vertical component v_{iy}.

5. A cabin is 2.60 km from a trailhead, in a direction 22.6° north of east. The trail runs due east along a river, then a side trail runs due north to the cabin.

a) How long is each section of trail?

b) Hikers walk the trails at 1.2 m/s, or they can bushwhack straight to the cabin through the forest at 0.80 m/s. Which route is faster, and by how many minutes?

6. Dr. Shafer drags a weighted training sled with a rope that makes 35.0° with the ground. The rope pulls with 420 N.

a) Find the horizontal and vertical components of the rope's pull.

b) Dr. Shafer says, "420 N at 35° gives me 344 N + 241 N = 585 N of pull." Explain the mistake.

c) Keeping the same 35.0° angle, how hard must the rope pull to give 400 N horizontally?

d) Dr. Shafer lowers the rope to 20.0° and pulls with 420 N again. Find the new components. Which one grew, and which one shrank?

7. Mr. White draws a soccer field with the passer at the origin, +x toward the opponent's goal and +y toward the left sideline. A teammate stands at $x = -18.0$ m, $y = +24.0$ m.

a) Find the magnitude and direction of the displacement from the passer to the teammate.

b) The pass rolls straight to the teammate at 12.0 m/s. Find v_x and v_y, with signs.

c) A defender stands at $x = -9.0$ m, $y = +10.0$ m. When the ball is 9.0 m in the −x direction from the passer, where is it in y? Does it pass the defender?

8. **Challenge.** Mr. Jantz walks city blocks, going only east and then north, to reach a spot a straight-line distance D away at angle θ north of east.

a) Show that the distance walked is $d = D(\cos\theta + \sin\theta)$. Use it for a coffee shop 650 m away at 30.0° N of E.

b) For a fixed D, what direction makes the walk longest? How many times longer than D is it?

c) A friend's house is 1.20 km away, and the street walk was 1.50 km. Find both possible directions to the house. (Hint: call the legs a and b. Desmos is fine.)